

Chapter 2: Kinetic Particle Theory

Matter

- Anything that has mass and occupies space
- Made up of particles called atoms, molecules or ~~ions~~ ions.

Kinetic Particle Theory

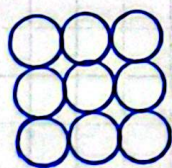
- Kinetic Particle Theory states that all matter is made up of tiny particles and these particles are in constant random motion.
- Kinetic Particle Theory explains the properties of the different states of matter.
- The particles in solids, liquids and gases have different amounts of energy. They are arranged differently and move in different ways.

Characteristics of Solids, Liquids and Gases

Solids

Arrangement of particles:

- Particles are very closely packed in an orderly manner.



Attractive forces:

- Very strong attractive forces between particles

Motion of particles:

- Particles rotate and vibrate about fixed positions.

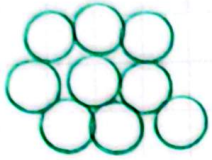
Energy:

- Particles possess very low kinetic energy.

Liquids

Arrangement of particles.

- Particles are closely packed in a disorderly manner.



Attractive forces:

- Strong attractive forces between particles.

Motion of particles:

- Particles slide over each other freely.

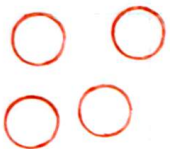
Energy:

- Particles possess low kinetic energy.

Gases

Arrangement of particles:

- Particles are spaced very far apart in a disorderly manner.



Attractive forces:

- Very weak attractive forces between particles.

Motion of particles:

- Particles move very quickly and randomly in any direction.

Energy:

- Particles possess very high kinetic energy.

Common Properties of solids, liquids, gases

Solids

Liquids

Gases

- | | | |
|------------------------|---|----------------------|
| - Cannot be compressed | - Cannot be compressed | - Can be compressed |
| - Have definite shape | - No definite shape | - No definite shape |
| - Have definite volume | - No takes the shape of containers | - No definite volume |
| | - Have definite volume | |

Changes of states

1. A change in state does not produce new substances.

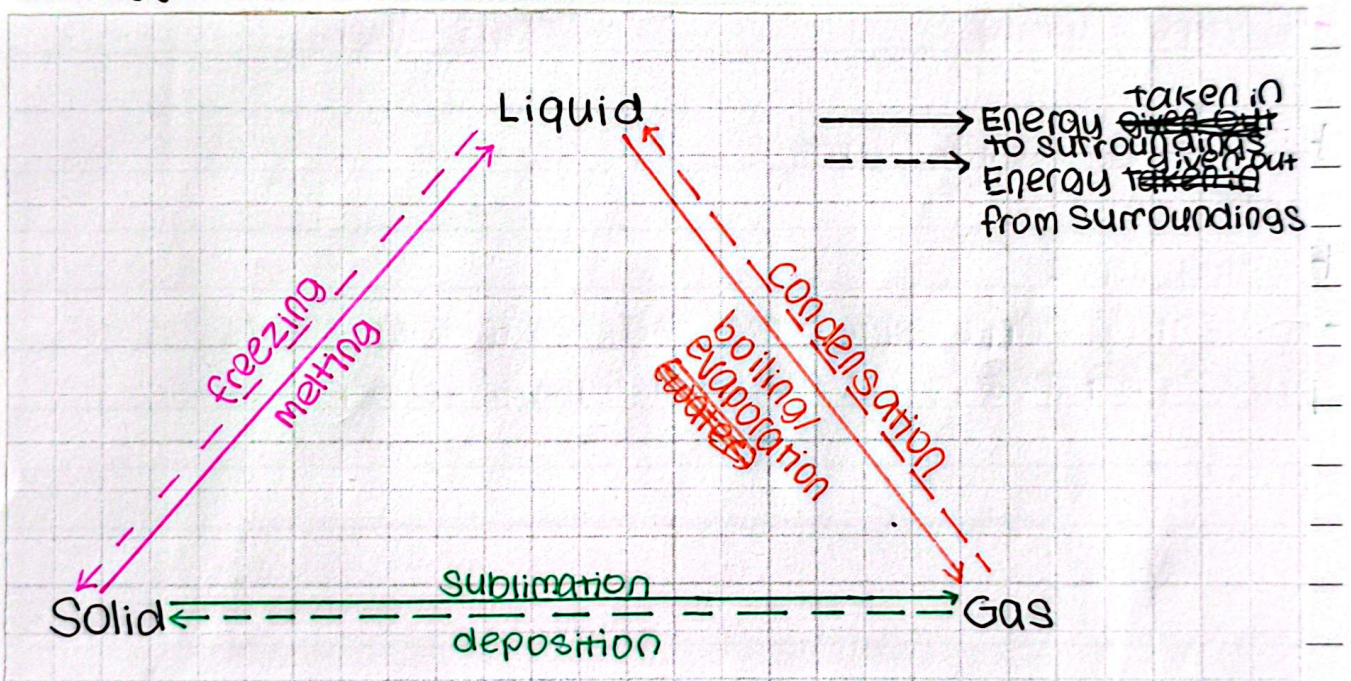
It is a physical change only.

~~2. Changes of state does not produce new substances.~~

2. Changes of state always occur with a change of energy.

Energy is either given out to the surroundings or taken in from the surroundings.

Energy changes during changes of states



Sublimation

- Change of state from solid to gas without melting.

Evaporation vs Boiling

- Evaporation is similar to boiling, but there are differences between them as well.
- Liquids that evaporate quickly at room temperature is called volatile liquids.

Boiling

- Occurs only at boiling temperature.
- Occurs throughout the liquid.
- Occurs rapidly.

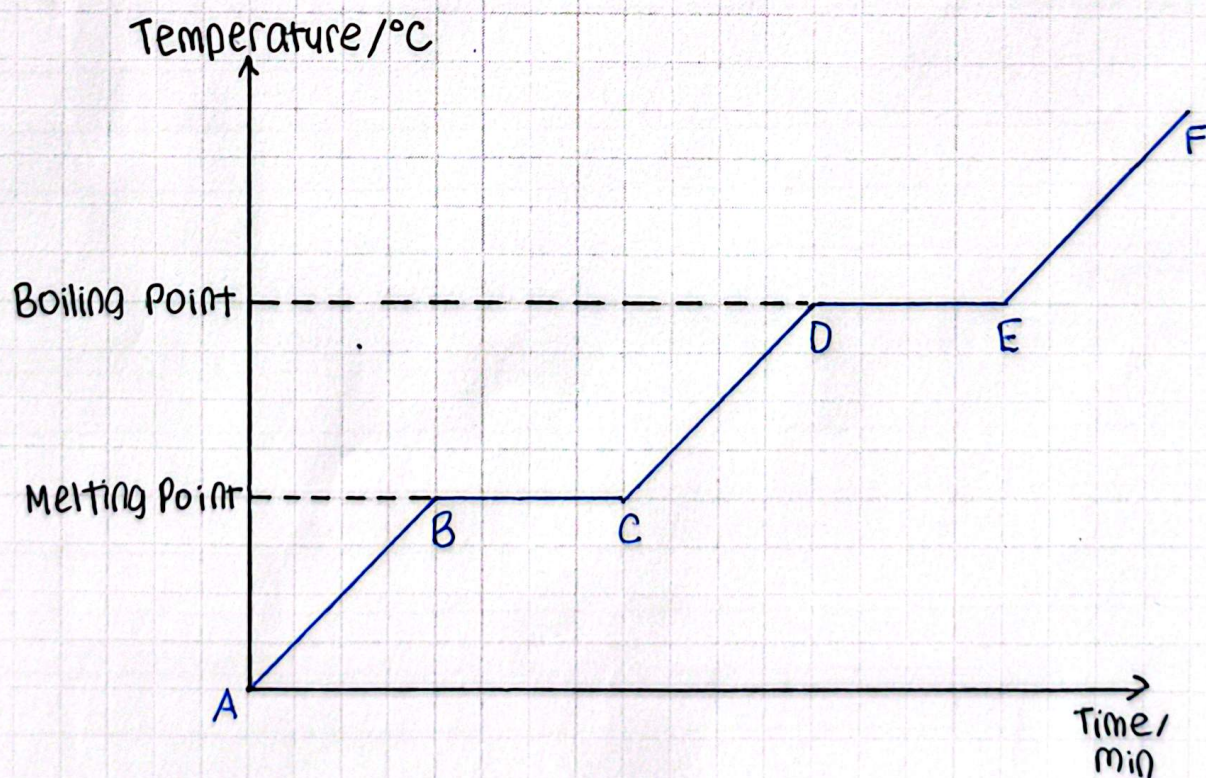
Evaporation

- Occurs at any temperature even below boiling temp.
- Occurs only at the surface of the liquid.
- Occurs slowly.

Heating and Cooling curves

Heating curve

- A heating curve shows the changes in temperature as a pure substance is heated from solid to liquid to gas.



Processes of the Heating Curve

1. A to B

- Substance is in solid state.
- As temperature increases, the particles gain energy & vibrate faster.

2. B to C

- Substance starts to melt as particles in the solid gain enough energy to overcome the very strong forces of attraction between them.

3. B to C

- Melting takes place. Mixture of solid and liquid.
- Temperature remains constant even as heating continues.
- All the heat taken in is used to overcome the attractive forces between particles. (and not to increase the temperature of the substance)

4. C

- All the substance has melted.
- Substance is in the liquid state.

5. C to D

- Substance is in liquid state.
- As the temperature increases, the particles gain energy and vibrate faster.

6. D

- Substance starts to boil as the particles in the liquid gain enough energy to overcome the strong forces of attraction between them.

7. D to E

- Boiling takes place. Mixture of liquid and gas.
- Temperature remains constant as all the ^{heat} energy taken in is used to overcome the attractive forces between particles.

8. E

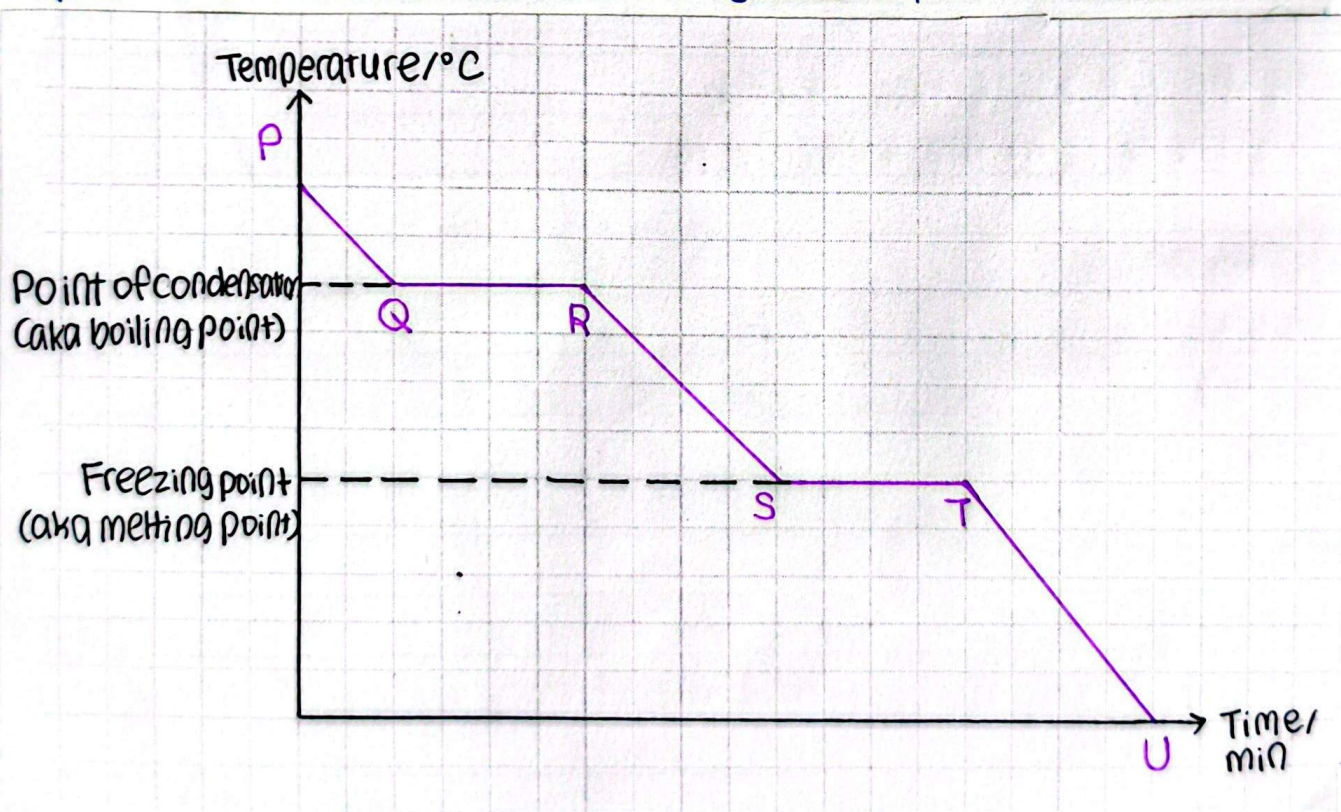
- All the substance has boiled. Thus the substance is now in the gaseous state.

9. E to F

- Substance is in the gaseous state. Particles gain energy and move faster.
- Temperature increases.

Cooling curve

- A cooling curve shows the changes in temperature as a pure substance is cooled from gas to liquid to solid.



Processes of the Cooling curve

1. P to Q

~~P to Q~~

- Substance is in gaseous state.
- As temperature decreases, particles lose energy and move about randomly at slower speed.

2. Q

- The substance starts to condense as particles in the gases lose energy and move slower and closer.
- The attractive forces between particles increases.

3. Q to R

- Condensation takes place. Substance exists as a mixture of liquid and gas.
- Temperature remains constant even as cooling continues. Heat is released as particles form strong attractive forces between each other.

4. R

- All the substance has condensed.
- The substance is now in the liquid state.

5. R to S

- As temperature decreases, the particles lose even more energy and move even slower and closer.

6. S

- Substance starts to freeze and as particles in the liquid lose energy and move slower and closer.
- The attractive forces between particles increases.

7. S to T

- Freezing takes place. Substance exists as a mixture of liquid and solid.
- Temperature remains constant even as cooling continues. Heat is released as particles form very strong attractive forces between each other.

8. T

- All the substance has frozen.
- The substance is now in the solid state.

9. T to U

- Temperature decreases.
- Particles lose energy and vibrate less vigorously.

Diffusion

- Diffusion is a phenomenon that shows that particles are in constant random motion.
- Diffusion is the movement of particles from a region of higher concentration to a region of lower concentration.

Rate of diffusion - Factors affecting rate of diffusion

1. Temperature

- The higher the temperature, the higher the kinetic energy of the particles, the faster the particles diffuse.

2. Molecular mass of particles

- The larger the molecular mass, the slower the rate of diffusion.

* Molecular mass refers to the mass of individual elements in the periodic table added together. Example: $H_2 \rightarrow \text{Mass} = 2$

Diffusion of gases - Rate of diffusion

- How to use chemical formula to calculate molecular mass

→ Use periodic table, under 'relative atomic mass'.

Example: Carbon monoxide, CO.

Mass of a carbon atom (C): 12

Mass of an oxygen atom (O): 16

Molecular mass of CO → $16 + 12 = 28$ → NO UNIT

↳ can be rewritten as M_r